Direct growth of crack-free stoichiometric silicon nitride on sapphire for wide-band photonics

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Low-loss stoichiometric silicon nitride (Si_3N_4) film on oxidized silicon has emerged as the major wide-band photonics platform for nonlinear and quantum photonics in the past decade. These films deposited by low-pressure chemical vapor deposition (LPCVD) are well known to have around 1 GPa residue tensile stress that often leads to cracking issues for film thickness above 300 nm. Though various stress mitigation approaches have been studied to isolate or prevent the cracking, these approaches put certain restrictions on the usage of Si_3N_4 and the stress issue remains a major limitation for foundry fabrication. On the other hand, low-residue-stress LPCVD silicon nitride on sapphire has been reported in 2018[1], suggesting sapphire is a much better substrate than silicon in hosting Si_3N_4. Unfortunately, this work has only been explored in photonic applications very recently at a telcom wavelength[2]. Here we advocate this platform with systematic data from material growth and stress analysis, and also optical measurements of the thin-film material and fabricated photonics devices across a broad range of wavelengths. We first carry out a comparison study of temperature-dependent stress analysis, and demonstrate direct growth of crack-free Si_3N_4 up to 2 micrometer with only 200 MPa stress. We provide index and loss measurements of the grown Si_3N_4 measured by prism coupling and ellipsometry. Our fabricated devices show a propagation loss of 0.35 dB/cm around 780 nm, 0.37 dB/cm around 1064 nm and 0.25 dB/cm at telecom, similar to the current results of Si_3N_4 on silicon. Our work highlights silicon nitride on sapphire as a promising wide-band photonics platform that further broaden the impact of silicon nitride photonics, particularly for operation at deep-visible and mid-infrared wavelengths, transparent substrate for vertical coupling and transmission, and thick Si_3N_4 film for direct wafer-scale fabrication.

I. INTRODUCTION

We advocate silicon nitride on sapphire[1,2] for wide-band photonics applications as an alternative to bypass silicon nitride on silicon's longstanding stress and growing issues and also with the promise to extended its performance for visible and infrared wavelengths. Silicon nitride on insulator, so far mainly on oxidized silicon wafers, has been the most extensively studied wide-band photonics platform to date[3] with photonic waveguides and resonators fabricated across a wide range of wavelengths across visible, near-infrared, and infrared spectra, as shown in Fig. 1(a). Silicon nitride has a wide transparency window reported from 400 nm to 6.7 um, with somewhat elevated loss ($>$ 2 dB/cm) in deep visible wavelengths[4], great at near-infrared ($\approx$ 0.2 dB/cm) and particularly excellent at telecom with concentrated research efforts ($<$ 0.1 dB/cm)[5], and also good above 3.5 um for mid-infrared where the major limits are its buffered silicon oxide layer[6]. A wide range of applications have been established in this low-loss photonic-integrated-circuit platform, encompassing ultra-high Q cavities using thick[7,8] or thin nitride[9,10], heterogeneous integration for active and electro-optical materials[11,12], and photonic quantum computing[13,14]. Silicon nitride has been the major wide-band photonics platform supporting nonlinear processes including octave-spanning frequency combs[15,16], widely-separated optical parametric oscillation[17,18], and photo-induced second harmonic generation[19,20]. Fig. 1(a) include silicon nitride with the two major deposition methods, low-pressure chemical vapor deposition (LPCVD) and plasmon-enhanced chemical vapor deposition (PECVD). We note that high confinement and low confinement Si_3N_4 devices can have even much lower loss values below 0.01 dB/cm, by either using ultra-thick Si_3N_4 typically with chemical mechanical polishing to avoid sidewall scattering, or using ultra-thin Si_3N_4 where the mode is mainly in the high quality oxide cladding than Si_3N_4. While both thick and thin nitrides can achieve ultra-low loss, and thin nitride is particularly useful in stimulated Brillouin scattering[9] and self-injection locking[10], most linear optical circuits[11–14] and wide-band nonlinear optical processes[15–20], including frequency comb generation, optical parametric oscillation, and harmonic generation, require moderately confined Si_3N_4 for a balance of dispersion engineering, coupling design, and optical qualities. We focus on moderately-confined Si_3N_4 in this work, whereas definition on thick, thin, and moderately-confined Si_3N_4 and their fabrication are discussed in a comprehensive review paper[5].

The major approach of silicon nitride photonics are growing stoichiometric silicon nitride by-low pressure chemical vapor deposition (LPCVD) on oxidized silicon wafers[42]. This approach can produce wafers in batch (for example, tens or hundreds of wafers per run), with relatively consistent results from furnace to furnace in different cleanrooms. When the gas ratio of ammonium gas and dichlorosilane gas is smaller than 2:1, the film becomes silicon rich and has low stress and no cracking issue, which is suitable for MEMS applications but exhibit excessive optical loss. When this gas ratio is above 2:1, the film becomes close to stoichiometric (a molecular ratio of Si:N = 3:4) with refractive indices still affected by specific gas ratios[43]. However, it is well known that the residue tensile stress of Si_3N_4 film is high, and often leads to cracking

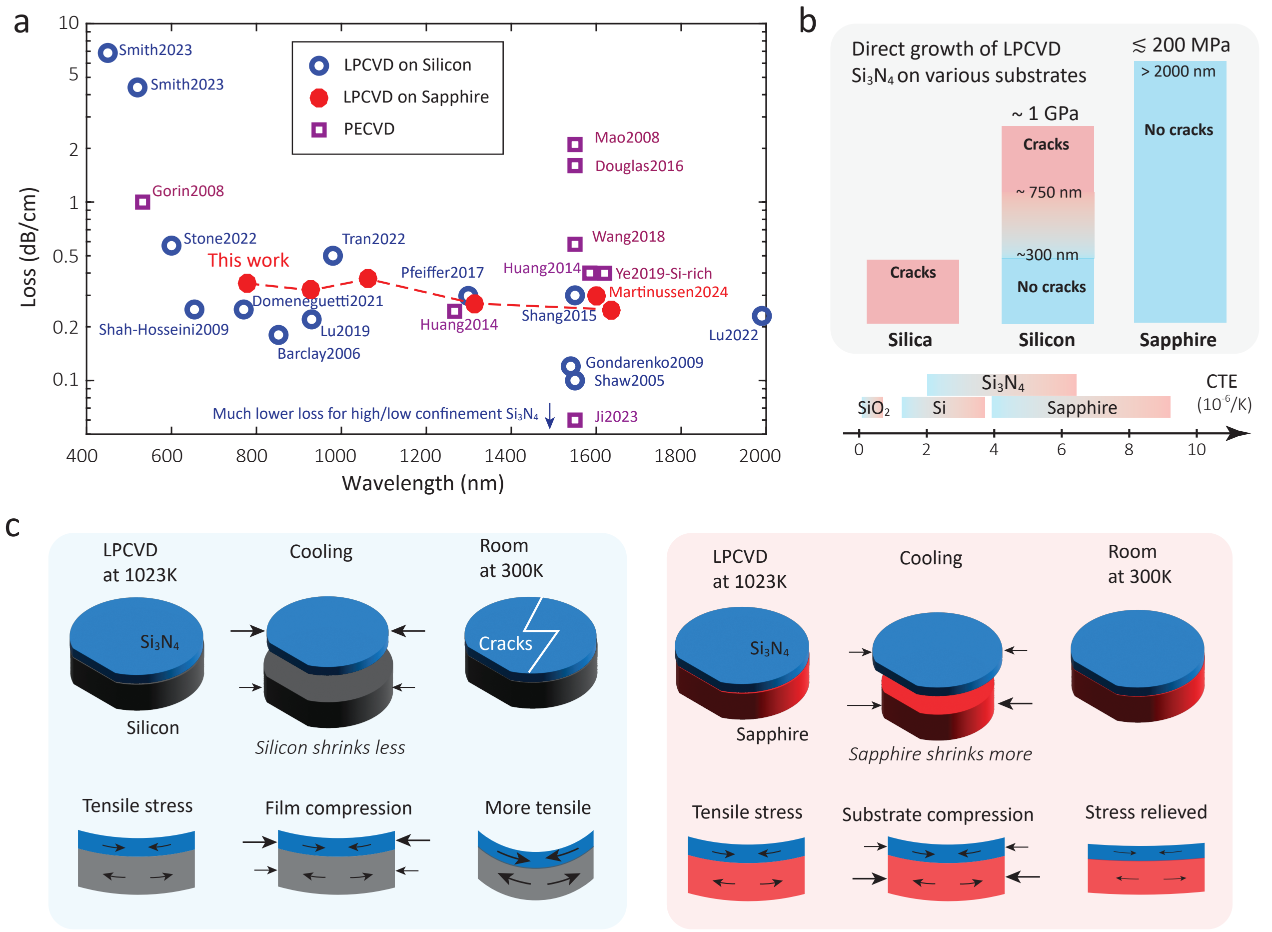

FIG. 1: **Comparison of stoichiometric low-pressure chemical vapor deposition (LPCVD) silicon nitride (Si_3N_4) on oxidized silicon wafers with plasmon-enhanced chemical vapor deposition (PECVD), emphasizing its broad applications, and different residue stress with different substrates. a,** The stoichiometric LPCVD Si_3N_4 is the most used method for wide band photonics applications (blue circles)[4,17,18,21–29]. The second widely used silicon nitride is by PECVD growth (red squares)[30–36]. Result from LPCVD Si_3N_4 on sapphire wafers (including this work) are in red circles[2]. **b,** The LPCVD Si_3N_4 growth has an unique stress problem, with around 1 GHz tensile stress. We note that in silica substrate with smaller coefficient of thermal expansion (CTE) than Si_3N_4, growth of crack-free Si_3N_4 film has not been reported, while in sapphire substrate with a larger CTE than Si_3N_4, the film is crack-free up to 2 μm reported in this work. **c,** The cracking in Si_3N_4 growth on silicon wafers typically happens not during the LPCVD growth at elevated temperature (1023 K to 1123 K), but during or after the cooling process to room temperature. The major reason is that the smaller CTE of Si, though closer to Si_3N_4's CTE, creates extra compression that leads to more residue tensile stress (left). The sapphire substrate can relieve the residue tensile stress (right) because of its larger CTE than Si_3N_4, so the growed film won't crack at all.

for films in unpatterned wafers.

Another approach used widely is plasmon-enhanced chemical vapor deposition (PECVD) [Fig. 1(b)]. PECVD silicon nitride has been reported with a focus on telecom wavelengths and CMOS-compatible fabrication, however, the materials from various facilities/tools have propagation losses reported over a wide range[2,31–36], and the material indices vary a lot depending on precursors and tools used, and annealing conditions used, and even the repeatibility from run to run is not guaranteed. One advantage is that PECVD can grow thicker film above 400 nm without stress issues, and also the film can be deposited in 300 nm wafers directly, both challenging for LPCVD. Other growth methods, including radio-frequency magnetron sputtering[44], atomic-pressure chemical vapor deposition[45], and molecular beam epitaxy (MBE)[46], are so far less promising in scaling than LPCVD and PECVD.

As discussed, most photonic applications focus on using LPCVD Si_3N_4 films, however, a serious problem with the stoichiometric LPCVD silicon nitride film is that it has a residue tensile stress of 900-1100 MPa at room temperature, which tends to form cracks across the wafer when the film is above 400 nm. Depending on furnace conditions and recipes, films above 400 nm may be deposited without cracks, but are extremely susceptible to cracking under any stress by mechanical and chemical perturbation, even small forces from normal wafer handling or chemical processing. This stress is a major limitation during the fabrication process, where dicing is almost impossible, chemical processes are quite limited, and

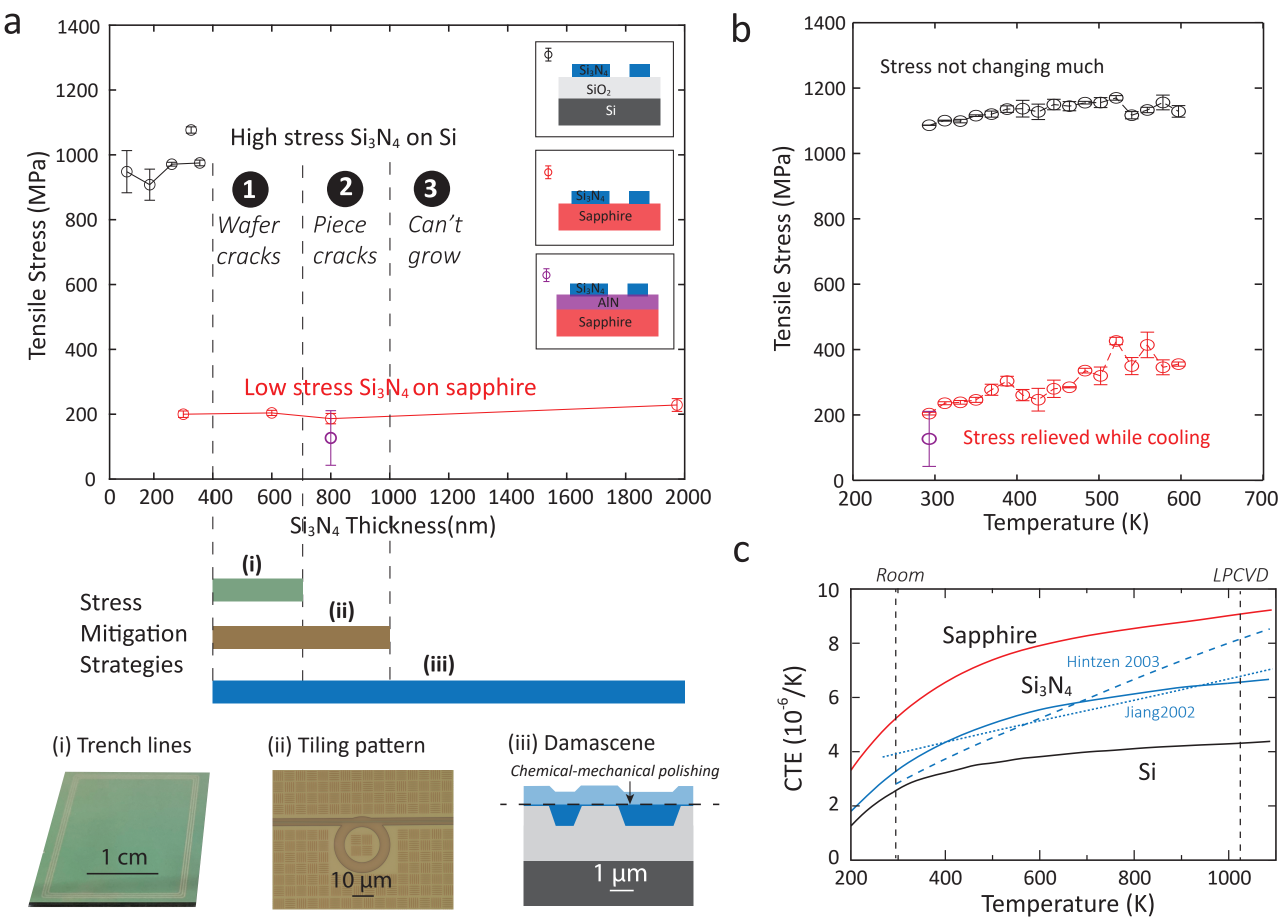

FIG. 2: **Stress characterization of SiN on Sapphire substrate (SiNoSa) compared to SiN on Silicon substrate (SiNoSi). a,** SiNoSa has much lower stress, and can grow up to at least 2 um; while SiNoSi cracks easily above 350-400 nm in wafer, above 600 nm for piece, and very challenging grow at all above 1000 nm. Three substrate conditions are shown to the right. Four risk mitigation approaches working for films with different thickness are shown at the bottom. Trench lines works for centimeter size with film thickness from 400 nm to 700 nm. Tiling pattern uses features of 10 to 100 μm filling the empty spaces throughout the wafer/chips. Damascene allows even thicker film to be grown in the micrometer scale both in depth (up to 2 micrometer) and size of the waveguide (up to a few micrometers) and followed by chemical mechanical polishing, and is often used together with tie pattern for crack terminations. **b,** Temperature-dependent stress measurement of SiNoSa versus SiNoSi. **c,** CTE of sapphire[37], Si_3N_4[38–40], and silicon[41] summarized, the data for silicon nitride from 1000 K to 1075 K are extrapolated. The theoretical data[40] and experimental data[38,39] roughly agree, and CTE data of Si_3N_4 are not as high quality or consistent in comparison to sapphire and silicon, largely due to the variation of the material. The error bars come from one-standard deviation value from measurements of stress at six rotational angles 30 degrees in step.

wafer spinning and drying needs manual operation and additional care. Various studies including trench lines before[47], in between of (for example, two growth each 300 nm), or after film growths (for films $<$ 400 nm), tiling patterns, rotate and grow[48], and damascene[49] have been explored as stress mitigation approach with a thickness up to 2.3 μm demonstrated[50], summarized at the bottom of Fig. 2(a). However, all these approaches have only relieved the stress in certain critical dimensions, but not reduced the stress universally in the deposited film in unpatterned wafers, and many fabrication limitations are associated with these processes. They all require pre-process during the wafer growth, thus increase the complication and also decrease the cleanliness of wafers and final device yields, and more or less creates complications for large-scale photonics foundry fabrication.

Most work are reported for Si_3N_4 on silicon substrates. We first take a step back and summarize the LPCVD growth of Si_3N_4 on various substrates, for example, fused silica, silicon, and sapphire, as illustrated in Fig. 1(b). It appears counter-intuitive that while silicon has the closest coefficient of thermal expansion (CTE) with silicon nitride, it is considerably worse than sapphire in terms of stress and cracks. Another important but less studied reference is fused silica, where we found it is challenging to grow even 100 nm Si_3N_4 on without cracking, even if we did trench lines. When we look at these three substrates and their outcomes, it supports the claim that Si_3N_4 favors the substrate with larger CTEs, and we will provide further study and data in Section II and III. The principle idea behind this claim is illustrated in Fig. 3, whereas substrate compression during cooling reduces the residue tensile

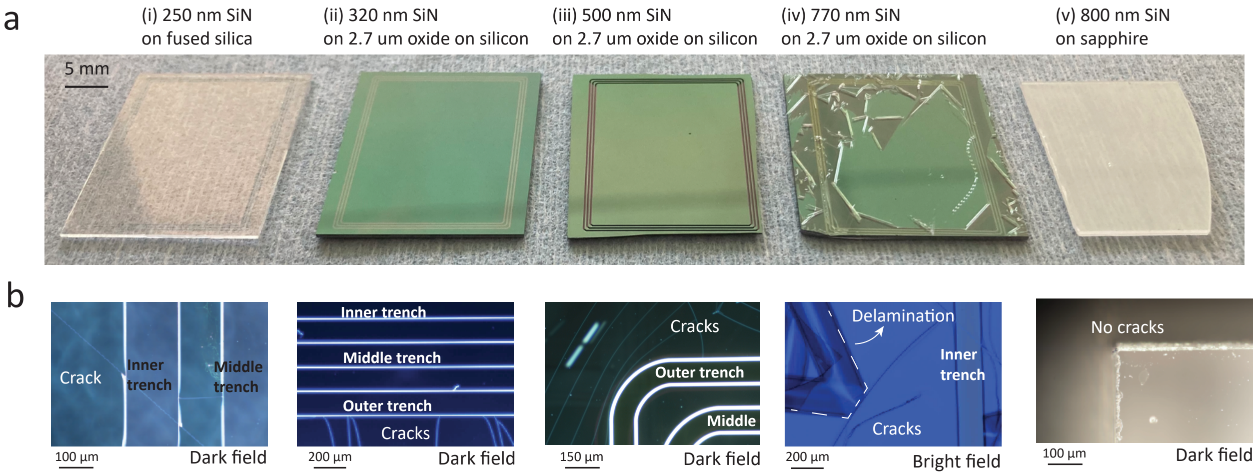

FIG. 3: **Cracking analysis of stoichiometric LPCVD SiN grown on different substrates. a,** Five chips on a cleanroom paper, with different thickness of LPCVD silicon nitride on fused silica (i), oxidized silicon (ii-iv), and sapphire (v). The furnace growth parameters stay the same. The inside area (i-iv) is around 1.6 cm x 2.1 cm. **b,** Fused silica, silicon, and sapphire substrates lead to complete different cracking phenomenon. Brightfield and darkfield microscope images showing the cracking inspection. (i) Cracking can often develop after deposition before etching trench lines and dicing. (ii,iii) Cracking are often stopped by the outside trench. (iv) The film has some cracks directly after deposition, but gets much worse after a few hours, with more cracks and even severe film delamination. (v) No cracks even near the diced edges, so no trench lines are needed. In (i-iv) the trenches are formed by three boxes with each trench 160 micrometers wide. **c,** A guidance for growing stoichiometric LPCVD SiN on these three substrate, considering trench lines are used for silica and silicon.

stress of the growed film. For Si_3N_4 growth on silicon substrate (left), when the LPCVD film is deposited, there is internal tensile stress at elevated temperature (typically 1023 k to 1123 K). During the cooling process, silicon has a smaller CTE than Si_3N_4 and therefore shrink less. This compress the Si_3N_4 film and make the residue stress more tensile, leading to more bowing of the wafer and cracks on the film that is thicker than 350 nm. Even when the film is not cracking, any external stress, like handling of metal tweezers, chemical process like RCA cleaning, mechanical process like spinning, can crack or even shattle the film. One evidence is that cracking almost always happen after the growth during or after cooling, not at the growth. On the other hand, sapphire is very good in hosting Si_3N_4 (right). At elevated temperature, Si_3N_4 is still deposited with tensile stress. However, during the cooling process, because sapphire shrinks more, while both substrate and film compresses more, the overall residue stress is relieved and much less tensile. As a result the Si_3N_4 film grown on sapphire is free from cracks, and can go through chemical and mechanical process like any low-stress Si-rich silicon nitride on silicon, though being stoichiometric Si_3N_4.

II. STRESS ANALYSIS

In this section, we show stress measurement of the wafers that supports the stress relieving hypothesis in Fig. 1(c).We compare mainly the Si_3N_4 grown on silicon and sapphire substrates in the same furnace run. The bottom line shows tensile stress of 900 MPa to 1100 MPa for the Si_3N_4 on silicon substrate up to 380 nm. The gas ratio is $NH_3 : DCS = 5 : 1$ to ensure stoichiometric growth[43] at a temperature of 1048 K and a pressure of 300 Pa. Above 400 nm the film starts to crack and stress measurement cannot be performed. Another wafer growth shows tensile stress exceeding 1000 MPa, and we use that wafer for temperature-dependent stress measurement later. Typically we need to do trench lines for thickness above 400 nm, tie pattern for film above 700 nm, and damascene for even thicker films. While the data we have shown here is Si_3N_4 grown on 2.7 μm silicon dioxide on silicon wafer, with the silicon dioxide formed by wet oxidation, the direct growth of Si_3N_4 on silicon shows similar tensile stress.

Figure 3(a) shows pieces of Si_3N_4 grown on different substrates. These pieces are about 1.6 cm to 2.2 cm in sizes. In fused silica, it is difficult to prevent silicon nitride to crack even when you have trench lines. In silicon substrate, the trench lines work well from 300 nm to 600 nm, while occasionally cracks can penetrate depending on the specific etching condition of lines. When the film is above 700 nm, the Si_3N_4 film always cracks, and start to delaminate after cooling down from growth temperature and few hours more afterwards in the cleanroom. In contrast, the Si_3N_4 grown on sapphire doesn't need trench lines, and show no cracks even close to diced edges.

We are able to grow up to 1960 nm without any cracks, as shown Fig. 2(a) (red). The stress is measured to be 200 MPa and shows no increasing of the growing thickness. Moreover, we have grown 800 nm Si_3N_4 with aluminum nitride on sapphire with success (purple). The aluminum nitride we use is 400 nm epitaxy grown on sapphire. Though data not shown here, we have also grown Si_3N_4 on 2 μm PECVD silicon oxide on sapphire with similar stress.

So far the stress measurements of Si_3N_4 have conclusively shown that sapphire substrate (with a much larger CTE than Si_3N_4), is advantageous in reducing tensile stress compared to

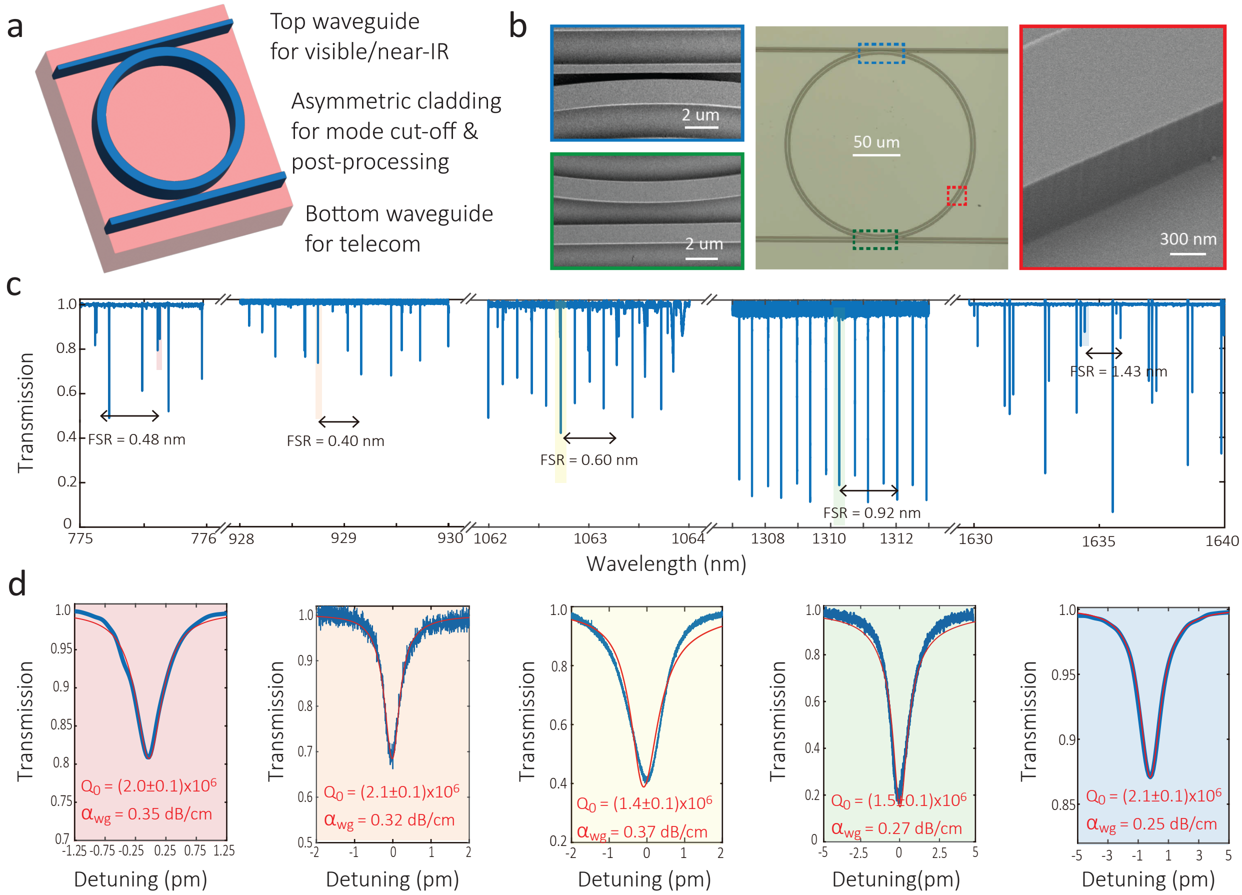

FIG. 4: **Fabricated photonic device and its characterization. a,** A microring (center, optical microscope image) with two waveguides coupling (left, scanning electron microscope images) from top and bottom for different wavelengths. The etched sidewall is quite straight and smooth (the right scanning electron microscope image). This device has a ring radius of 150 μm, a ring width of 1.5 μm, and a thickness of 650 nm. **b,** The wavelength-dependent refractive indices extracted by ellipsometry. The Si_3N_4 on sapphire shows slightly smaller refractive indices on sapphire substrate, with AlN (blue dashed line) buffered layer or not (red solid line), than grown on oxidized silicon substrate (black). **c d e,** The measured optical transmission trace shows a good number of optical resonances with reasonable coupling around 775nm,1050nm and 1630nm. **f,** Three fundamental transverse-electric modes at different bands are fitted with intrinsic optical quality factors of $(2.4\pm0.1)\times10^6$, $(1.4\pm0.1)\times10^6$, and $(1.3\pm0.1)\times10^6$. The uncertainties come from one-standard deviation of the nonlinear fittings.

silicon substrate (with a close but relatively smaller CTE than Si_3N_4), as well as fused silica substrate (smallest CTE of the three). While Fig. 1(c) seems to be a possible explanation, it is important to examine the stress data with temperature variation. Because of the technical constraints, we cannot readily measure the stress while the wafers are cooling down from 1023 K, but we can measure the stress heating them up from temperature to 600 K. The tool we use is xx, and it is measuring residue stress by the bowing of the wafer. We find that the residue stress of the silicon wafer stays more or less consistently high, and the stress of the sapphire wafer is much higher than heated up, increasing from 200 MPa to 360-400 MPa at 500-600 K.

III. PHOTONIC CHARACTERIZATION

We fabricate Si_3N_4 device with a piece of 1.8x2.2 cm. The piece is diced from a 4 inch wafer, with 620 nm Si_3N_4 grown on the sapphire substrate directly. Figure 4(a) shows the optical microscope image of the device (center), the scanning electron microscope (SEM) images of the top and bottom coupling waveguides (left) for visible/near-infrared and telecom wavelengths, respectively, and the SEM image of the etched sidewall. The sidewall smoothness is at similar levels, with a good etch stop because sapphire is very resistive to flourine etching. The etching rate is $<$ 1 nm per minute for sapphire, $\approx$ 30 nm per minute for Si_3N_4, and $>$ 20 nm per minute for SiO_2. The etching is the same recipe we used for Si_3N_4 on silicon substrate, using reactive ion etching with CHF_3:O_2 gases at flow rates of 30:2 sccm and power of 150 W at a pressure of 15 mTorr. The selectivity ratio of Si_3N_4:ZEP520a (electron beam lithography resist) and Si_3N_4:DUV210 (deep ultraviolet photoresist) is around 1.3:1 and 1.5:1.

We measure the wavelength-dependent refractive indices by Woolam ellipsometer xx. The Si_3N_4 on sapphire is found to be slightly smaller than that is grown on silicon. Si_3N_4 on AlN on sapphire shows similar indices to that directly grown on sapphire at shorter wavelengths, but closer to grown on

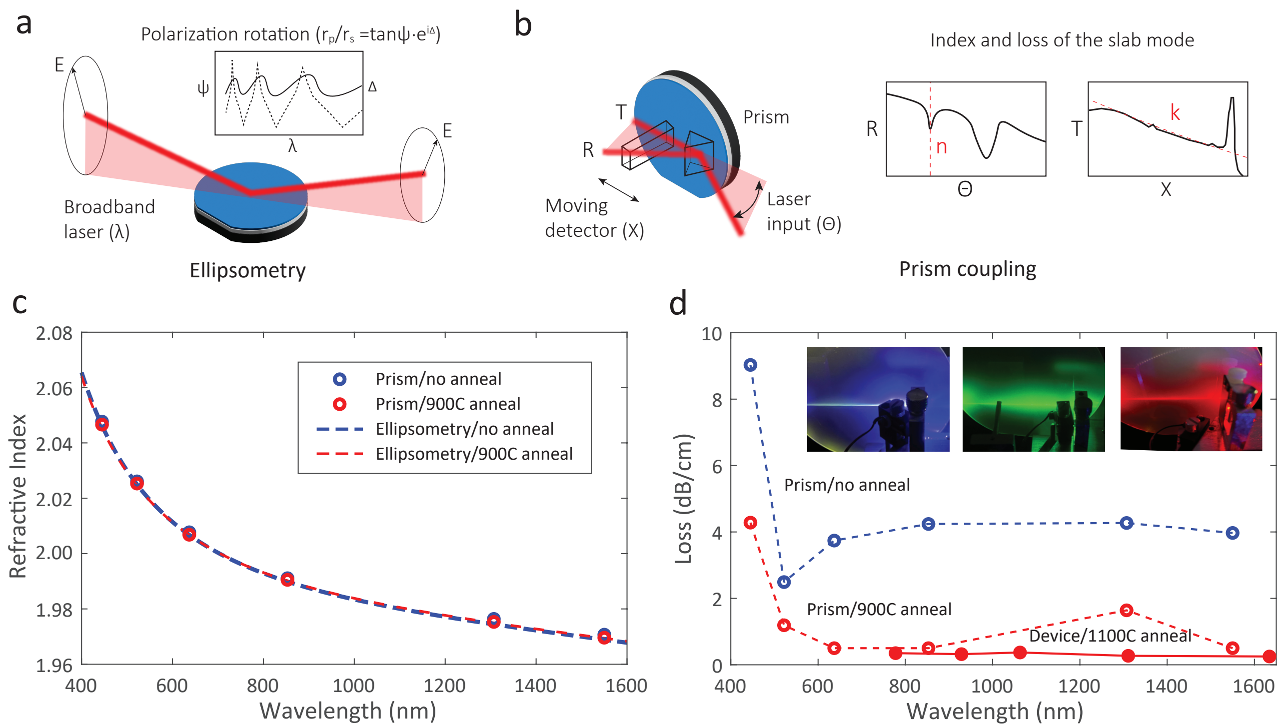

FIG. 5: **Ellipsometry and metricon measurement of SiNOS wafers before and after annealing. a,** Images showing the Metricon measurements at blue, green, and red. **b,** Refractive index and loss of thin film characterized by ellipsometry (solid lines) and metricon (open circles) before and after annealing. **c,** Measured refractive indices by prism coupling (circles) and ellipsometry (lines). **d,** Measured loss by prism coupling (dashed line) and measured devices (solid line).

silicon at telecom. However, because AlN is a birefringent material, and this difference could come from the uncertainty of fitting because of the added layer and/or the slight change of material property of AlN. Further study needs to be carried out with various AlN wafers, perhaps with annealing at elevated temperatures first.

We measure the cavity transmission by lens fiber coupling using lasers at different colors. At 775 nm, we find the fundamental transverse-electric mode has an intrinsic optical quality of $(2.4\pm0.1)\times10^6$, in a microring with a radius of 150 μm, a ring width of 1625 nm, and a free spectral range of 0.48 nm. The propagation loss is thus calculated to be (0.19 ± 0.01) dB/cm at 775 nm. The intrinsic quality factor measured at 1062 nm is $(1.4\pm0.1)\times10^6$, the loss is (0.25 ± 0.02) dB/cm. Then its highest intrinsic quality factor for the U band is $(1.3\pm0.1)\times10^6$, which corresponding to (0.25 ± 0.02) dB/cm. This value is similar to our previous work using similar recipe, but with Si_3N_4 on oxidized silicon wafer (Fig. 1(a)).

We characterize the thin film by both ellipsometry and prism coupling, as illustrated by Fig. 5(a,b). The refractive indices measured by both methods are consistent with each other 5(c). The un-annealed and annealed films have refractive indices of $n^2 = 1 + 2.850/[1-(0.13451/\lambda)^2] - 0.01623\lambda^2$ and $n^2 = 1 + 2.853/[1-(0.13229/\lambda)^2] - 0.01281\lambda^2$, respectively.

The film annealed at 900 degree Celsius shows smaller optical loss than un-annealed film in Fig. 5(d). We note that it is challenging to characterize losses below 0.5 dB/cm by prism coupling in our Metricon tool, though in principle a smaller loss can be extracted according to tool specification. Our devices are annealed at a higher temperature at 1100 degree Celsius, and showing losses below 0.5 dB/cm, where we also include in Fig. 1(a) to compare with other results.

IV. CONCLUSIONS

We grown crack-free stoichiometric silicon nitride film without any stress mitigation approaches, with up to 2 μm film on sapphire substrate. We show stress data for different thickness of films, and also temperature-dependent measurements of stress, in support of using higher CTE substrate for silicon nitride growth. We further show its advantage in photonics fabrication, and show low optical losses in par with current silicon nitride on silicon. The results at visible seems to be promising, which is in line with its relatively smaller refractive indices, suggesting the film is more nitrogen rich. We also investigate asymetric cladding for device protection with multi-wavelength ring-waveguide coupling, and also symmetric cladding for fiber-chip coupling, as well as preliminary packaging work along this end. Overall we hope to provide sufficient information to advocate silicon nitride on sapphire as a promising wide-band photonics platform, particularly wafer-scale fabrication for deep visible and mid-infrared wavelengths.

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